lecture notes 10

Gravitational Self Force

This is a totally different approximation route. The idea is to expand in mass ratio $q = m_2/m_1$. This is most useful when $q \to 0$ and a very small object is inspiraling into a much more massive one. We usually call these **extreme mass-ratio inspirals**. They spend many years at $v \lesssim c$, so PN doesn't work! Numerically very hard also.

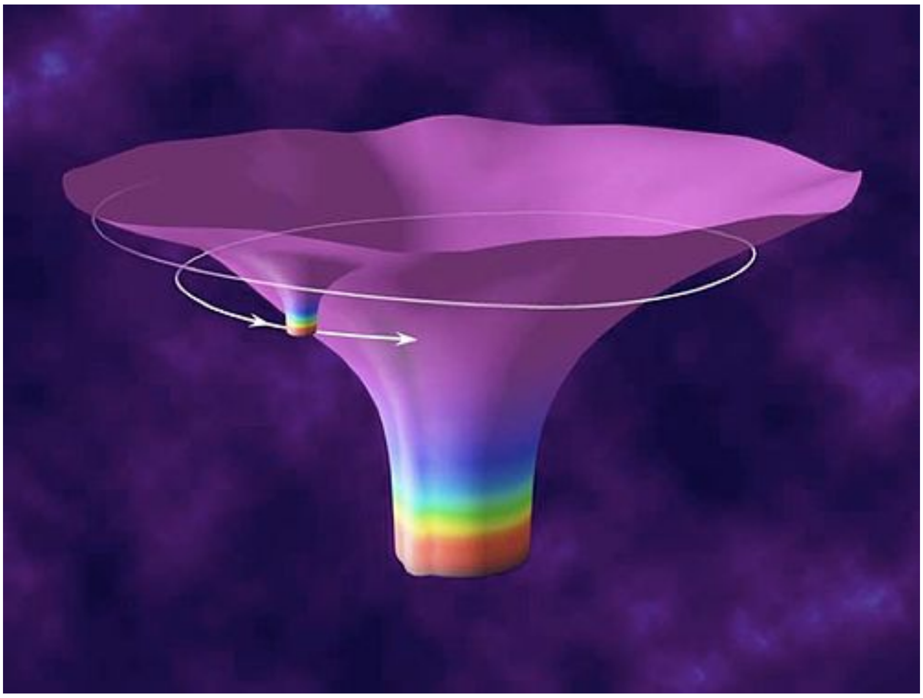


Figure 1: 786f681f1d7f8bc23e4517ce767afa17.png

The basic idea: it's easy to calculate geodesics of massless objects in the big mass's spacetime. These might not be simple, but it is easy to calculate them, just solve the geodesic equation.

Rotating black holes (in isolation) follow the Kerr metric:

$$ds^2 = -(1-\frac{r_s r}{\Sigma})dt^2+\frac{\Sigma}{\Delta}dr^2+\Sigma d\theta^2+\left(r^2+a^2+\frac{r_s r a^2}{\Sigma}\sin^2\theta\right)\sin^2\theta d\phi^2-\frac{2r_s r a\sin^2\theta}{\Sigma}dtd\phi$$

And you guys remember the geodesic equation:

$$\ddot{x}^\mu + \Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$$

So, just calculate the Christoffel symbols and you have an equation for the geodesics. They're not very easy to solve for analytically. But they look pretty cool:

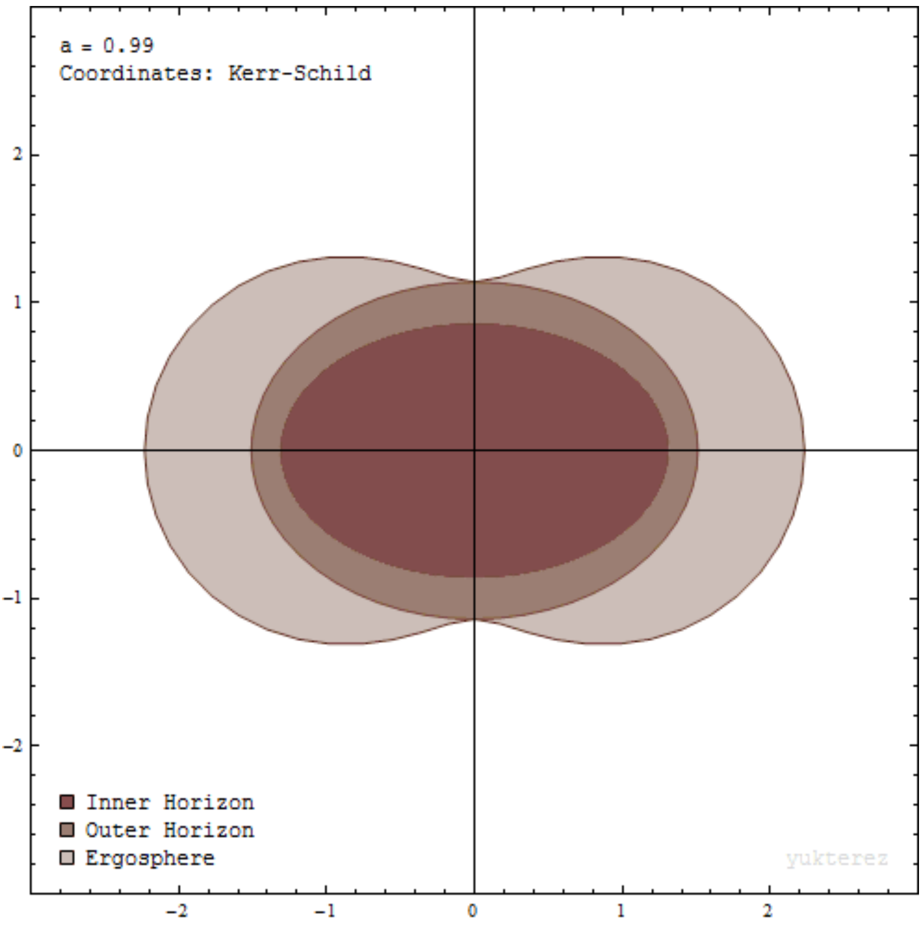


Figure 2: Ergosphere_and_event_horizons_of_a_rotating_black_hole.gif

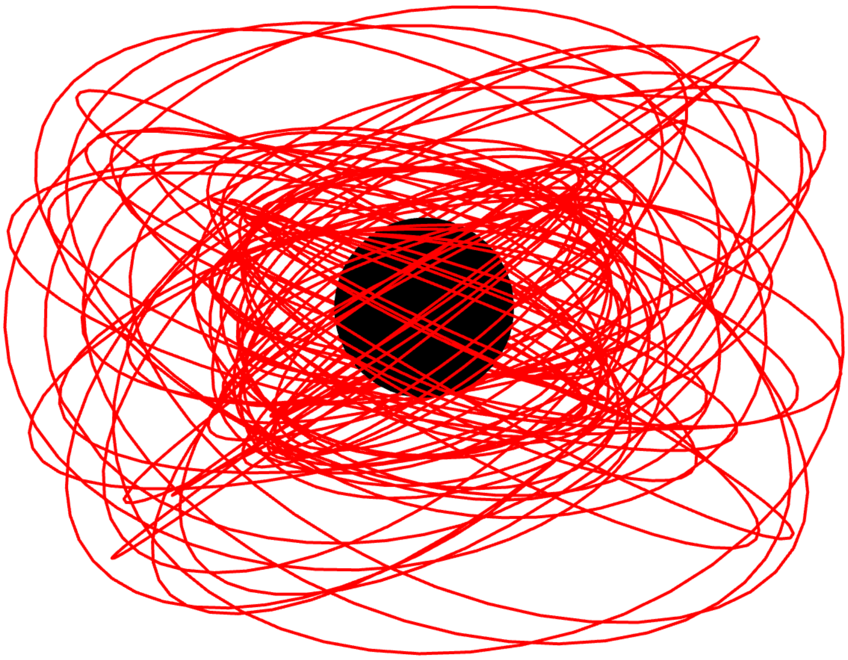


Figure 3: 6d897254038175d10a7a2f639b7318e6.png

For the self-force approximation, we want to expand the metric with small corrections due to the mass of the secondary:

$$g_{\mu\nu}^{\text{eff}} = g_{\mu\nu}^{\text{primary}} + q h_{\mu\nu}^{(1)} + q^2 h_{\mu\nu}^{(2)} + \mathcal{O}(q^3)$$

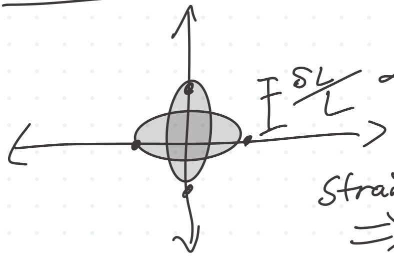
The secondary's path will be a geodesic in this effective spacetime. This is different than a geodesic in the primary's metric, so it looks like there's an extra force on it. "self force"

This is not an easy calculation and needs to be done on a computer. Nowadays we have results up to second order (2-GSF). These work surprisingly well even up to $q \to 1$.

Another nice thing is these are fully relativistic the whole time: most of the technicalities of the PN expansion don't apply :)

Lots of these results are available here if you're curious: https://bhptoolkit.org/documentation.html

Gravitational Wave Detectors



$\frac{\delta L}{L} \propto h$

Strain (h) is a fractional change in L.
$\Rightarrow$ Bigger L, bigger δL.

Very long L high-precision

EM: Detectors typically count photons (intensity)

$$I \propto |E|^2 + |B|^2 \propto 1/d^2$$

GW: We can measure strain directly!

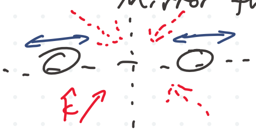
$$\boxed{h \propto 1/d}$$

(no movement)

If you know the intrinsic amplitude of the source, you can figure out the sky location based on the observed amplitude.

4 spots on the sky.

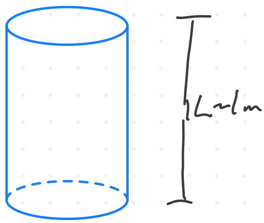
Mirror this scenario in vertical and horizontal.



GW detectors are quadrupolar.

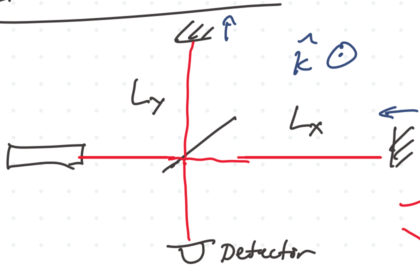
Kinds of detectors

- Rulers
  - Resonant bars (Weber bars)
  - Interferometers
- Timing
  - Pulsar timing arrays



GWs create acoustic oscillations in the bar.

Interferometers



Strongest response when $f_{GW} \approx f_{resonance}$

Destructive interference when $L_x = L_y + n\lambda$, $n \in \mathbb{Z}$

Constructive interference max. when $L_x = L_y + n\lambda/2$

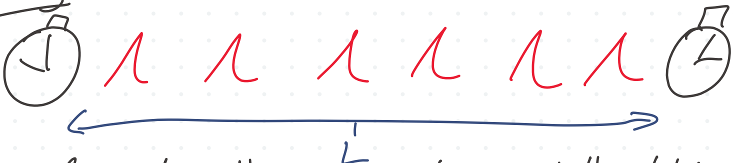
Laser wavelength also gets stretched when a GW passes.
$h = 0 \longrightarrow h > 0$, light stretches,
but time in interferometer $\ll$ time of detecting a GW
$\mathcal{O}(10\ \text{km}/c)$ $\sim 1\,\text{s}$
$\sim 10^{-5}\,\text{s}$

$$h \sim \delta L / L$$

$$\delta L \sim h\,\lambda$$

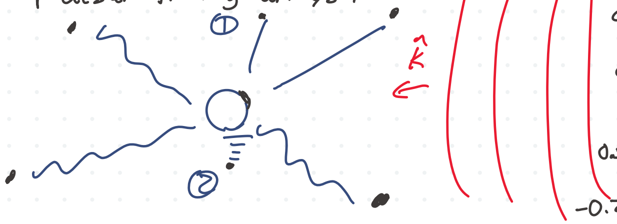
$\mathcal{O}(10^{-30}\,\text{m})$ $\mathcal{O}(10^{-21})$ ↳ small, $\mathcal{O}(\text{nm})$

Timing

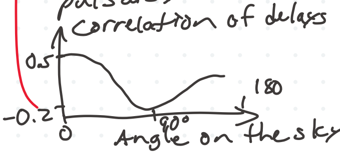


If you know the emission time and the delay $\Rightarrow \delta L$.
Shifts in time-of-arrival.

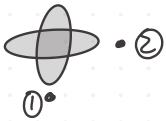
Pulsar timing arrays:



We don't know L of the pulsars, but we can compare correlations between pulsars.



1 + 2 will be anti-correlated



Hellings-Down curve